HCML Monitoring Portal: ERD by domain

Admin & system

PORTAL_USER			
string	name	PK	parsed from PORTAL_USERS, name:password:role
string	pass		plain text or scrypt hash
string	role		viewer, operator or admin

STATS		
number	hosts	
number	items	
number	triggers	
number	groups	
number	problems	
number	unacknowledged	
string	bySeverity	counts keyed 0 to 5

AI layer

CHAT_TURN		
string	role	user or assistant
string	content	capped at 2000 chars, last 12 turns kept

a fresh snapshot per user message

CHAT_SNAPSHOT			
number	generatedAt	PK	one per assistant message, never stored
number	hosts		
number	sites		
number	problems		
number	unacknowledged		
number	slas		
number	degradedServices		
boolean	truncated		a cap fired: the model sees a PARTIAL estate
string	text		plain text, ~4 chars per token, capped at 5500

PROBLEM_EXPLANATION		
string	summary	schema-enforced, never empty
string	businessImpact	
string	recommendation	

one entry per input tag, none invented

TAG_EXPLAINED		
string	tag	
string	value	
string	meaning	what the tag tells a non-engineer

SLA_EXPLANATION		
string	status	a short verdict
string	plain	the acronyms removed
boolean	meetingTarget	
string	recommendation	

up to 20 degraded services

up to 10 enabled SLAs

derives: humanize() through an LLM

Services & SLA

ZBX_SLA · sla			
string	slaid	PK	
string	name		
string	slo		target percent
string	period		0 daily, 1 weekly, 2 monthly, 3 quarterly, 4 annually
string	status		0 disabled, 1 enabled
string	timezone		
string	description		

sla.getsli, one row per service

ZBX_SLI · computed, no table		
number	sli	achieved percent
number	uptime	seconds
number	downtime	seconds
number	error_budget	seconds, negative = SLO missed

POSITIONAL index, not a key

ZBX_SERVICE · services			
string	serviceid	PK	
string	name		
number	status		-1 OK, else the severity that propagated up
string	algorithm		carried through, never mapped to a label
string	sortorder		
string	description		

selectChildren: a DAG, not a tree

selectTags

ZBX_SERVICE_TAG · service_tag	
string	tag
string	value

derives: up to 40 problems, worst first

derives: SLI grafted onto the node

derives: humanize() through an LLM

derives: the DAG materialised

SERVICE_NODE			
string	serviceid	PK	
string	name		
number	status		-1 OK, else severity
number	worst		max status anywhere in the subtree
number	descendants		DISTINCT ids below: the DAG can share children

children, DAG-safe descendant count

first SLA wins if several apply

SERVICE_SLA			
string	slaid	PK	
string	name		
number	slo		
number	sli		
number	errorBudget		seconds, negative = missed
boolean	meeting		sli >= slo

selectProblemEvents, the root causes

Events & Incidents

ZBX_EVENT · problem, events			
string	eventid	PK	
string	objectid	FK	the trigger id, no constraint in MySQL
string	object		0 = trigger, the only kind handled
string	source		0 = triggers, always sent never read
string	value		1 = PROBLEM
string	name		the notification text
string	severity		0 to 5
string	clock		unix seconds
string	r_eventid	FK	0 means STILL OPEN
string	acknowledged		0 or 1
string	opdata		the live value
string	suppressed		declared, never branched on

r_eventid, the recovery row

objectid, only when object=0

selectTags

derives: paired with its recovery row

derives: unacknowledged, bucketed by age

derives: counted by objectid, top 100

ZBX_TRIGGER · triggers			
string	triggerid	PK	
string	manual_close		1 = the Close action is permitted

ZBX_EVENT_TAG · problem_tag	
string	tag
string	value

INCIDENT			
string	eventid	PK	
string	objectid	FK	trigger
string	hostid	FK	
string	name		
string	severity		
number	start		clipped to the report window
number	end		recovery clock, or now when still open
boolean	resolved		
boolean	acknowledged		

AGING_BUCKET			
string	label	PK	Under 1h, 1-4h, 4-24h, Over 24h
number	count		

TOP_TRIGGER			
string	objectid	PK	
string	name		taken from the EVENT, never from the trigger
string	severity		
string	host		
number	count		

derives: grouped by objectid

NOISY_TRIGGER			
string	objectid	PK	
string	name		
string	host		
string	severity		
number	count		
number	medianDuration		MEDIAN, not mean: one outlier must not hide many
number	shortLived		
number	ackRate		0 to 1
number	totalDuration		
number	longest		
boolean	stillOpen		
string	flags		flapping, unactioned, chronic

up to 30 sites, worst first

selectHosts: FIRST HOST ONLY

derives: overlapping intervals merged per host

Hosts & Inventory

ZBX_HOST · hosts			
string	hostid	PK	
string	host		technical name, read only for map labels
string	name		the visible name
string	status		0 monitored, 1 not monitored
string	maintenance_status		1 = in maintenance
string	description		

HOST_AVAILABILITY			
string	hostid	PK	
string	host		
number	availability		percent, overlapping problems merged
number	downtime		seconds
number	incidents		
number	longest		

selectHostGroups / selectHosts

selectTags

selectInventory, [] when disabled

selectInterfaces

ZBX_HOST_TAG · host_tag		
string	tag	site, owner, criticality
string	value	

ZBX_HOST_INVENTORY · host_inventory		
string	site_city	site signal, rank 2
string	location	site signal, rank 3
string	poc_1_name	owner signal
string	poc_1_email	owner signal
string	notes	requested but never read

ZBX_HOST_INTERFACE · interface		
string	ip	
string	dns	map labels only
string	useip	1 = connect by IP
string	main	1 = the default interface
string	type	1 agent, 2 SNMP, 3 IPMI, 4 JMX
string	available	0 unknown, 1 up, 2 down

ZBX_HOST_GROUP · hstgrp			
string	groupid	PK	
string	name		site fallback when no tag or inventory

derives: open problems and worst severity per host

derives: severity histogram per group

derives: compliance rolled up per group

GROUP_PROBLEMS			
string	groupid	PK	
string	name		
number	total		
string	bySeverity		counts keyed 0 to 5

GROUP_SCORE			
string	name	PK	the host group name
number	hosts		a host counts in EVERY group it belongs to
number	complete		
number	pct		

derives: only when a dimension is missing

hostids

elements[0].hostid, host elements only

Governance scorecard

SCORECARD_DIMENSION			
string	key	PK	naming, site, owner, criticality
string	label		
string	hint		
number	present		
number	total		
number	pct		
boolean	scored		false when the rule is not configured

a gap names the dimensions it misses

HOST_GAP			
string	hostid	PK	
string	name		
string	site		
string	missing		which of the four dimensions are absent

Metrics & Items

ZBX_ITEM · items			
string	itemid	PK	
string	key_		parsed for icmpping target and mode
string	name		
string	value_type		0 float, 1 char, 2 log, 3 uint, 4 text
string	units		
string	lastvalue		not a column, taken from history
string	lastclock		blank beyond the history period, 24h default
string	prevvalue		not a column, taken from history
string	state		0 normal, 1 not supported, in item_rtdata

derives: resolveSite() tag, name, inventory, group

itemids, table picked by value_type

selectTags, link_group + link_role

itemids, hourly aggregates

derives: trend.get, history fallback

ZBX_HISTORY · five history tables			
string	itemid	FK	no constraint in MySQL
string	clock		the newest one anchors a graph window
string	value		

ZBX_ITEM_TAG · item_tag		
string	tag	link_group, link_role
string	value	

ZBX_TREND · trends, trends_uint			
string	itemid	FK	no constraint in MySQL
string	value_avg		
string	value_max		

CAPACITY_ROW			
string	itemid	PK	
string	host		
string	metric		cpu, memory or disk
string	units		
number	avg		
number	max		
string	source		trend, history or none

Maps

ZBX_MAP · sysmaps		
string	sysmapid	PK
string	name	
string	width	
string	height	

selectLinks

ZBX_MAP_LINK · sysmaps_links			
string	linkid	PK	
string	selementid1	FK	
string	selementid2	FK	
string	color		hex without the #

selectSelements

selementid1 and selementid2

ZBX_MAP_SELEMENT · sysmaps_elements			
string	selementid	PK	
string	label		
string	x		
string	y		
string	elementtype		0 host, 1 map, 2 trigger, 3 group, 4 image
string	hostid	FK	elements[0], host elements only

derives: labels resolved, host status attached

MAP_ELEMENT			
string	selementid	PK	
string	labelText		HOSTNAME and HOST.IP macros resolved
string	hostid	FK	host elements only
string	hostName		
number	problems		open problems on that host
number	maxSeverity		worst of them, -1 when clear

derives: 1..3 icmpping items, keyed hostid:target

Sites & Links

SITE			
string	name	PK	derived, never stored anywhere
number	total		
number	available		
number	unavailable		
number	unknown		
number	maintenance		
number	disabled		
number	problems		
number	unacknowledged		
number	worst		highest severity here, -1 when clear

a total function, every host lands in one site

LINK_PATH			
string	name	PK	the link_group value
string	state		DOWN only when every leg is down

grouped by the link_group tag

SITE_HOST			
string	hostid	PK	
string	name		
string	availability		worst-wins across the host interfaces
string	siteSource		tag, name, inventory or group
number	problems		

LINK			
string	id	PK	hostid:target, a synthetic composite key
string	host		
string	target		ping destination, parsed from the item key
string	label		taken from the icmppingloss item name
boolean	up		from icmpping
number	loss		percent, from icmppingloss
number	latency		ms, from icmppingsec
number	jitter		rttMax minus rttMin, needs both modes
string	state		up, degraded, down, unknown
string	group		link_group tag, absent = in no path
string	role		link_role tag, e.g. main or standby
string	lastclock		